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Vessel for receiving a stereolithographic resin, a device at which a stereolithographic object is made, a method for making a stereolithographic object and a method for making a vessel for receiving a stereolithographic resin

Abstract

A vessel for receiving a stereolithographic resin that includes an oxygen permeable element comprising a stereolithographic resin receiving surface that is an interior vessel surface. The oxygen permeable element comprises at least one marginal surface separate from the stereolithographic resin receiving surface in fluid communication with a gas comprising oxygen for ingress of the oxygen and transport thereof to the stereolithographic resin receiving surface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0046072	12/2015	Rolland	264/401	B29C 64/124
2016/0082671	12/2015	Joyce	425/174.4	B29C 64/307
2017/0113417	12/2016	Deotte	N/A	B29C 64/124
2018/0264724	12/2017	Feller	N/A	B29C 64/393
2018/0304526	12/2017	Feller	N/A	B33Y 40/20
2018/0370136	12/2017	Stadlmann	N/A	N/A
2020/0094468	12/2019	Feller	N/A	B29C 64/124
2020/0171746	12/2019	Schmidt	N/A	B29C 64/245

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 2018/187709	12/2017	WO	N/A
WO 2019/217325	12/2018	WO	N/A

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Background/Summary

PRIORITY CLAIM

(1) The present application is a National Phase entry of PCT Application No. PCT/AU2020/050231, filed Mar. 12, 2020, which claims the benefit of Australian Patent Application No. 2019900819, filed Mar. 12, 2019, which are incorporated herein by reference, in their entireties.

FIELD OF THE INVENTION

(2) The present invention generally relates to a vessel for receiving a stereolithographic resin, a device at which a stereolithographic object is made, a method for making a stereolithographic object, and a method for making a vessel for receiving a stereolithographic resin.

BACKGROUND OF THE INVENTION

(3) A three dimensional object can be built up one section at a time (i.e. layer wise). A layer of stereolithographic resin is solidified in the shape of a section of the object. Once the section is formed, another is formed in contact with the previous section. Repetition of this process allows multi-laminate objects in the form of stereolithographic objects to be fabricated. A stereolithographic object made by step-wise layer fabrication may be a rapid prototype, or finished product, for example.

(4) A solidified layer of stereolithographic resin may, however, adhere to a surface on which it is made. A step of detaching the adhered solidified layer of stereolithographic resin from the surface may be required. It may be desirable to prevent the solidified layer of stereolithographic resin being formed adhered to the surface on which it is made, which may increase the rate of fabrication and may reduce separation forces that may otherwise damage the object being formed. Oxygenation of a layer of stereolithographic resin at the surface may prevent solidification of the layer, which may prevent the solidified layer of stereolithographic resin being formed adhered to the surface. Generally, however, the introduction of a stream of pure oxygen has been required for oxygenation of the layer of stereolithographic resin at the surface.

SUMMARY OF INVENTION

(5) Disclosed herein is a vessel for receiving a stereolithographic resin. The vessel comprises an oxygen permeable element comprising a stereolithographic resin receiving surface that is an interior vessel surface. The oxygen permeable element comprises at least one marginal surface separate from the stereolithographic resin receiving surface for fluid communication with a gas comprising oxygen for ingress of the oxygen and transport thereof to the stereolithographic resin receiving surface.

(6) In an embodiment, the at least one marginal surface is in fluid communication with an atmosphere for passive ingress of the oxygen.

(7) In an embodiment, the combined area of the at least one marginal surface is greater than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, and 32% of the area of the stereolithographic resin receiving surface. The combined area of the at least one marginal surface may be less than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, and 32% of the area of the stereolithographic resin receiving surface.

(8) In an embodiment, the oxygen permeable element has a thickness greater than at least one of 1 mm, 2 mm, 4 mm, 8 mm, 16 mm, 32 mm, 64 mm and 128 mm. The oxygen permeable element may have a thickness less than at least one of 1 mm, 2 mm, 4 mm, 8 mm, 16 mm, 32 mm, 64 mm and 128 mm.

(9) In an embodiment, the oxygen permeable element has a thickness no less than one hundredth of the square root of the area of the stereolithographic resin receiving surface. The oxygen permeable element may have a thickness no more than one tenth of the square root of the area of the stereolithographic resin receiving surface.

(10) In an embodiment, the oxygen permeable element comprises a semipermeable coating at the stereolithographic resin receiving surface. The semipermeable coating may be permeable to oxygen but not the stereolithographic resin. The semipermeable coating may comprise at least one of an amorphous fluoropolymer, Teflon AF1600, Teflon AF2400, polyphenylene oxide, ethylcellulose, poly(1-trimethylsilyl-1-propyne), polymethylpentene, and poly(4-methyl-2-pentyne).

(11) In an embodiment, the semipermeable coating has a thickness in the range of 0.1 μm to 1000 μm. The semipermeable coating may have a thickness in the range of 1 μm to 20 μm. The semipermeable coating may have a thickness in the range of 5 μm to 10 μm. The semipermeable coating may have an oxygen permeability of no less than one of 10 Barrer, 30 Barrer, 1000 Barrer,

2000 Barrer and 9000 Barrer. The semipermeable coating may have an oxygen permeability of no more than one of 20 Barrer, 50 Barrer, 2000 Barrer, 3000 Barrer and 10 000 Barrer. The oxygen permeable coating may have the structure of an oxygen permeable coating formed by evaporative casting. The oxygen permeable element may comprise a oxygen permeable substrate. The oxygen permeable substrate may have a sheet configuration. The semipermeable coating may be deposited on the oxygen permeable substrate. The oxygen permeable substrate may comprise at least one of polydimethylsiloxane, poly(4-methyl-2-pentyne) and polymethylpentene. The oxygen permeable substrate may have an oxygen permeability of at least 550 barrer. The oxygen permeable substrate may have a Shore A hardness of 10-100 Durometer. The oxygen permeable element may comprise at least one of polydimethylsiloxane, poly(4-methyl-2-pentyne) and polymethylpentene.
(12) In an embodiment, the at least one marginal surface comprises at least one edge surface.
(13) In an embodiment, the oxygen permeable element may be transparent to an actinic radiation for hardening the stereolithographic resin.
(14) In an embodiment, an exterior vessel surface comprises the at least one marginal surface.
(15) In an embodiment, an exterior vessel surface is perforated. The exterior vessel surface may be perforated adjacent the at least one marginal surface. The marginal surface may be in fluid communication with the gas via perforations in the exterior vessel surface. The exterior vessel surface may have a perforation area greater than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, 32% and 64% of the area of the stereolithographic resin receiving surface. A perforation in the exterior vessel surface may be covered by an oxygen permeable sheet.
(16) In an embodiment, the vessel comprises at least one side wall perforated adjacent the at least one marginal surface.
(17) In an embodiment, the marginal surface comprises an edge surface.
(18) An embodiment comprises a wall supporting the oxygen permeable element. The wall may comprise an actinic radiation transparent material. The actinic radiation transparent material may comprise at least one of a glass, a polymer and poly (methyl methacrylate). The actinic radiation transparent material may have a Young's modulus of 1 GPa to 100 GPa.
(19) Disclosed herein is a device at which a stereolithographic object is made. The device comprises a vessel for receiving a stereolithographic resin in accordance with the above disclosure.
(20) An embodiment comprises a radiation source configured to irradiate the stereolithographic resin receiving area with a radiation The radiation source may be a light source. The radiation may be an actinic light.
(21) An embodiment comprises a radiation manipulator configured to manipulate the radiation. The radiation manipulator may impart a spatial feature to the radiation. The radiation manipulator may impart a temporal feature to the radiation. The shape of each section formed may thus be individually controlled by the action of the radiation manipulator on the light.
(22) In the context of this specification, a section is to be understood to encompass a slice of the stereolithographic object. A planar section encompasses a portion of the stereolithographic object located between two parallel planes that intersect the stereolithographic object. Generally, but not necessarily, the sections formed are planar sections.
(23) In an embodiment, the radiation manipulator is configured to scan the radiation relative to the surface.
(24) Disclosed herein is a method for making a stereolithographic object. The method comprises the step of disposing a stereolithographic resin on a stereolithographic resin receiving surface of an oxygen permeable element The oxygen permeable element comprises at least one marginal surface separate from the stereolithographic resin receiving surface. The at least one marginal surface is in fluid communication with a source of oxygen for ingress of oxygen and transport thereof. The method comprises forming a layer of oxygenated stereolithography resin in contact with the stereolithographic resin receiving surface by the transport of the oxygen. The method comprises the step of irradiating the stereolithographic resin so disposed with an actinic radiation to selectively

harden a layer of stereolithographic resin adjacent the layer of oxygenated stereolithographic resin. The method comprises the step of separating the selectively hardened layer of stereolithographic resin and the layer of oxygenated stereolithographic resin.
(25) In an embodiment, the at least one marginal surface is in fluid communication with atmosphere for passive ingress of atmospheric oxygen. Forming the layer of oxygenated stereolithography resin may comprise transport of the atmospheric oxygen.
(26) In an embodiment, the vessel is in accordance with the above disclosure.
(27) In an embodiment, the layer of selectively hardened stereolithographic resin constitutes a section.
(28) Disclosed herein is a method for making a vessel for receiving a stereolithographic resin. The method comprises the step of forming within a vessel scaffold an oxygen permeable element comprising a stereolithographic resin receiving surface that is an interior vessel surface. The oxygen permeable element comprises at least one marginal surface separate from the stereolithographic resin receiving surface. The at least one marginal surface is for passive ingress of oxygen and transport thereof to the stereolithographic resin receiving surface.
(29) In an embodiment the step of forming within the vessel scaffold the oxygen permeable element comprises the step of forming an oxygen permeable substrate in the vessel shell. The step of forming within the vessel shell the oxygen permeable element may comprise forming a semipermeable coating on a surface of the oxygen permeable substrate.
(30) In an embodiment, the vessel is in accordance with the above disclosure.
(31) The stereolithographic object may be fabricated by sequentially irradiating each of a plurality of layers of stereolithographic resin to form respective sections. Each section may be an entire section of the stereolithographic object. Each section may be formed spaced apart from the stereolithographic resin receiving surface by an oxygenated layer of stereolithographic resin. Each entire section may comprise an entire planar section.
(32) Irradiation may also attach the solidified material to the stereolithographic object being made.
(33) Especially delicate stereolithographic objects may be made that may not be made using another approach.
(34) Were possible, any one or more features of above disclosure may be combined with any one or more features of the above disclosure as is suitable.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) In order to achieve a better understanding of the nature of the present invention, embodiments will now be described, by way of example only, with reference to the accompanying figures in which:
(2) FIG. **1** shows an orthographic view of one example of a vessel scaffold of an embodiment of a vessel for receiving a stereolithographic resin for making a stereolithographic object.
(3) FIG. **2** shows a section through an embodiment of the vessel comprising the vessel scaffold of FIG. **1**.
(4) FIG. **3** shows another section through the embodiment of the vessel of FIG. **2**.
(5) FIG. **4** shows the section through the embodiment of the vessel of FIG. **2** having received therein stereolithographic resin and illustrating the transport of atmospheric oxygen to the stereolithographic resin.
(6) FIGS. **5** to **7** and **9** show schematic elevation views of an embodiment of a device making a stereolithographic object, and illustrate steps of an embodiment of a method for making a stereolithographic object.
(7) FIG. **8** shows a detail of FIG. **7**.

(8) FIGS. **10** to **12** show schematic views of example radiation sources that may form part of a device for making a stereolithographic object of FIGS. **5** to **7** and **9**;
(9) FIG. **13** shows an example architecture of a controller for controlling the devices of the preceding figures;
(10) FIGS. **14** and **15** show sections through alternative embodiments of a vessel for receiving a stereolithographic resin for making a stereolithographic object.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

(11) FIG. **1** shows an example of a vessel scaffold generally indicated by numeral **105** in the form of a frame or shell, the vessel scaffold **1-5** being for a vessel **100** for receiving a stereolithographic resin **104**. An object in the form of a stereolithographic object can be made at the vessel **100**. The vessel scaffold **105** comprises at least one perimeter exterior wall **106** with perforations in the form of gas ports **194***a*-**194***p* ("**194**") cut through the wall **106**. The gas ports **194** may be covered with an oxygen permeable sheet, for example a mesh, paper or generally any suitable oxygen permeable membrane as suitable or desired.

(12) FIG. **2** shows a section through the vessel **100**, the section corresponding to section A-A indicated in FIG. **1**. The vessel **100** comprises a vessel wall **101** in the form of a structural member. In this embodiment, the vessel wall is a vessel bottom wall **101**, however in alternative embodiments it may be a top wall or side wall, for example. The vessel **100** comprises an oxygen permeable element **210**, which has substantially a rectangular prism or sheet configuration. The oxygen permeable element **210** comprises a stereolithographic resin receiving surface **204** that is an interior bottom vessel surface, however it may for example be an interior side vessel surface. In the context of the present document, an interior vessel surface is a surface of a vessel **100** that is substantially or entirely associated with the inside of the vessel **100**. The oxygen permeable element **210** comprises at least one marginal surface **206** in the form of an edge surface at the exterior of the vessel and which is separate from the stereolithographic resin receiving surface **204**. The marginal surface **206** is in fluid communication with a source of oxygen in the form of atmosphere **208** for passive ingress of atmospheric oxygen and transport thereof to the stereolithographic resin receiving surface **204**. In the context of the present document, atmosphere **208** is the gas enveloping the vessel **100** (or device comprising the vessel), and which comprises oxygen ("atmospheric oxygen"). An example of a suitable atmosphere includes the earth's atmosphere or air taken from the earth's atmosphere, which comprises approximately 21% oxygen. In alternative embodiments, the marginal surface may be on the same face as the stereolithographic resin receiving surface **204**, or on the underside, such as surface **213** of FIG. **15**.

(13) In this but not necessarily in all embodiments, the oxygen permeable element **210** comprises a semipermeable coating **103** and a oxygen permeable substrate **102**. The oxygen permeable element **210** is a laminate, however an alternative embodiment may be without a coating and have a homogenous structure and/or composition. In the context of the present application, "semipermeable" encompasses selectively permeable. The semipermeable coating **103** is permeable to oxygen but not the stereolithographic resin. Vessel bottom wall **101**, and the oxygen permeable element **210** are transparent to actinic radiation in the form of actinic light used in the stereolithographic process to harden stereolithographic resin received on the stereolithographic resin receiving surface **204**. Other embodiments may not comprise semipermeable coating **103**.

(14) Vessel bottom wall **101** provides rigidity and mechanical support for the oxygen permeable element **210**. In the present embodiment, the vessel bottom wall **101** comprises a plate of an actinic radiation transparent material in the form of fused silica. The plate **101** is 6 mm thick, however it may have a lesser or greater thickness as suitable. The vessel bottom wall **101** may alternatively or additionally comprise B270 glass, a polymer such as polycarbonate, Perspex, PET or generally any suitable material. The thickness of vessel bottom wall **101** may be between 1 mm and 100 mm, for example, depending on the strength required by the apparatus, but other thicknesses may be suitable. The actinic radiation transparent material has a Young's modulus of 1 GPa to 100 GPa,

although it may be greater or lesser in other embodiments.
(15) Oxygen permeable element **210** permits passive transport of ambient oxygen from outside the vessel **100** via gas ports **194** at which the at least one marginal surface **206** is disposed.
(16) In the present embodiment, oxygen permeable substrate **102** is a 7 mm thick layer of platinum cured polydimethylsiloxane, such as Sylgard 184 or Elastosil RT601, however other suitable substrates materials of other thicknesses may be used. Vessel **100** was constructed by the applicant and has dimensions 170 mm×100 mm×6 mm (XYZ) with 12 gas ports **194** each of area 180 mm.sup.2 (30 mm×6 mm) comprising a 6 mm thick vessel wall **101** comprising a plate of low-iron glass on which the substrate **102** and a 10 μm layer of Teflon AF2400 was cast. Other dimensions and materials may be used.
(17) The substrate **102** is cast directly into the frame **106** onto the vessel bottom wall **101**. The oxygen permeable substrate **102** has a Shore A hardness of 10-100 Durometer, but other suitable harnesses may be used. Polydimethylsiloxane has a high oxygen permeability of around 600 Barrer. In other embodiments, the oxygen permeable substrate **102** may have generally any suitable oxygen permeability in accordance with the application. In alternative embodiments the thickness of the oxygen permeable element may be between 1 mm and 128 mm, however in some embodiments greater or lesser thickness may be used as appropriate, depending, for example, on the area of the vessel **100** in the XY plane. Larger areas may require higher fluence of oxygen into the vessel necessitating a greater thickness of oxygen permeable substrate **102**.
(18) The oxygen permeable coating **103** generally prevents the stereolithography resin or its substituent ingredients from diffusing into the substrate **102**, impairing its oxygen permeability or optical clarity. Oxygen permeable coating **103** is optional, however it may extend the working life of the vessel **100**. In the present embodiment, the oxygen permeable coating **103** comprises a 10 μm thick layer of amorphous fluoropolymer such as Teflon AF2400 or Teflon AF1600. The thickness of the coating **103** may be, for example, between 0.1 μm and 1000 μm in some embodiments, and 5 μm to 10 μm in other embodiments. Teflon AF2400 has a relatively high oxygen permeability of 1300 Barrer. The coating **103** is in the present but not all embodiments cast directly onto the top surface of substrate **102**. The Teflon AF2400 is dissolved in a suitable fluorinated solvent such as Fluorinert FC-40 from 3M or Galden HT-135 from Solvay Solexis to form a solute, then evaporating the solvent component of the solute to form the film coating **103**. Alternatively, a pre-made film of Teflon AF2400 or Teflon AF1600 may be bonded to the oxygen permeable substrate **102** using a polydimethylsiloxane or other adhesive. Alternative films for the oxygen permeable coating **103** include polymethylpentene (oxygen permeability 37 Barrer), polyphenylene oxide (Oxygen permeability 17 Barrer), ethylcellulose (oxygen permeability 11 Barrer), poly(1-trimethylsilyl-1-propyne) (oxygen permeability 9700 Barrer), or poly(4-methyl-2-pentyne) (oxygen permeability 2700 Barrer). A thinner layer of lower permeability film may provide a similar rate of oxygen transport to a thicker layer of a higher permeability film. The semipermeable coating **103** generally has an oxygen permeability of no less than one of 10 Barrer, 30 Barrer, 1000 Barrer, 2000 Barrer and 9000 Barrer. The semipermeable coating **103** has an oxygen permeability of generally no more than one of 20 Barrer, 50 Barrer, 2000 Barrer, 3000 Barrer and 10 000 Barrer. Other oxygen permeabilities may be used as suitable.
(19) FIG. **14** shows a section through another embodiment of a vessel **300** for receiving a stereolithographic resin for making a stereolithographic object, where parts that are similar or identical in form and/or function to those in FIGS. **1** and **2** are similarly numbered. In this embodiment, the gas port **194** comprises a conduit coupling **209** in the form of a threaded gas hose connector, or alternatively a nipple fitting or generally any suitable conduit coupling, for communicating oxygen from a supply of oxygen to the vessel **300**. The gas port **194** defines a chamber in the form of a pressure chamber. Pressurised gas in form of compressed air, a gas with a concentration of oxygen greater than that of earth's atmosphere, or pure oxygen, for example, may be supplied by a hose connected to the conduit coupling **209**, for example. The hose communicates

the gaseous output from an oxygen concentrator, however other oxygen sources may be used. For example, a pressurized oxygen cylinder may be connected to the gas port **194** via the gas hose to supply oxygen.

(20) FIG. **15** shows a section through another embodiment of a vessel **400** for receiving a stereolithographic resin for making a stereolithographic object, where parts that are similar or identical in form and/or function to those in FIGS. **1** and **2** are similarly numbered. In this embodiment, the gas port **212** comprises a conduit coupling in the form of a threaded gas hose connector, or alternatively a nipple fitting or generally any suitable conduit coupling, for communicating oxygen from a supply of oxygen to the vessel **400**. The gas port **212** couples to a chamber **211** in the form of a pressure chamber, which allows supply of gas to the lower surface **213** of oxygen permeable substrate **102**. Pressurised gas in form of compressed air, a gas with a concentration of oxygen greater than that of earth's atmosphere, or pure oxygen, for example, may be supplied by a hose connected to the gas port **212**, for example. The hose communicates the gaseous output from an oxygen concentrator, however other oxygen sources may be used. For example, a pressurized oxygen cylinder may be connected to the gas port **212** via the gas hose to supply oxygen to the oxygen permeable substrate **102**.

(21) FIG. **3** shows another section through the vessel **100** of FIG. **2**, the section corresponding to section B-B indicated in FIG. **1**. FIG. **3** illustrates the ratio of the area of the gas ports **194** to the area of the perimeter wall of vessel **100**. The gas ports **194** occupy the majority of the outer surface of the perimeter of the vessel **100** in the XY section occupied by the oxygen permeable element **210**. The perimeter wall **106** area is generally but not necessarily reduced to improve atmospheric oxygen flow through the gas ports **194** while maintaining the structural integrity of the wall. This arrangement presents a significant portion of the oxygen permeable element **210** to the ambient atmosphere (generally “air”) to encourage the flux of oxygen from outside the perimeter of the vessel to its center.

(22) FIG. **4** shows another schematic diagram of the vessel shown in FIG. **2**, having received stereolithographic resin **104**. In FIG. **4**, atmospheric oxygen, indicated generally by the numeral **900**, diffuses into the oxygen permeable element **210** via the gas ports **194**. The relative concentration of oxygen is conveyed graphically by the spacing of the illustrated particles. At the edge of the oxygen permeable element the concentration of oxygen particles is that of ambient atmospheric concentration, generally at substantially 1 atmosphere of pressure. In the present embodiment, there is no mechanical air or oxygen feed system to pressurize or artificially concentrate the oxygen particles or force them by contrived means of pressure or concentration gradient into the oxygen permeable element **210**. The diffusion process is an example of a passive process, in that it requires no externally applied motivation. Some oxygen molecules, such as that indicated by the numeral **901**, diffuse through the oxygen permeable member **210** and into the stereolithographic resin **104** received by the stereolithographic resin receiving surface **204**. The thickness of the oxygen permeable element **210** in the z-direction is generally but not necessarily selected with consideration of the area of stereolithographic resin receiving surface **204**. The greater the area of the stereolithographic resin receiving surface **204**, the thicker the element **210** may be. Trial and error experiments can be performed to determine whether the thickness is sufficient to form an oxygenated layer of stereolithographic resin that resists photohardening, in which progressively thicker elements **210** can be used. The oxygen permeable element **210** has a thickness greater than at least one of 1 mm, 2 mm, 4 mm, 8 mm, 16 mm, 32 mm, 64 mm and 128 mm. The oxygen permeable element **210** has a thickness less than at least one of 1 mm, 2 mm, 4 mm, 8 mm, 16 mm, 32 mm, 64 mm and 128 mm. Other thicknesses may be used as suitable and desired. The oxygen permeable element **210** generally but not necessarily has a thickness no less than one hundredth of the square root of the area of the stereolithographic resin receiving surface **204**, and no more than one tenth of the square root of the area of the stereolithographic resin receiving surface **204**, to enable sufficient oxygen diffusion.

(23) The combined area of the at least one marginal surface **206** is generally but not necessarily greater than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, 32%, and 100% of the area of the stereolithographic resin receiving surface **204**. The combined area of the marginal surface generally but not necessarily is less than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, 32% and 1000% of the area of the stereolithographic resin receiving surface **204**.
(24) The exterior vessel surface generally but not necessarily has a perforation area (that is, the combined area of all the gas ports **194**) that is greater than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, 32% and 64% of the area of the stereolithographic resin receiving surface **204**.
(25) The dimension of the gas ports **194** in the z-direction can be chosen to be substantially the thickness of the oxygen permeable element **210**, to expose substantially all of the area to ambient atmosphere without the stereolithographic resin **104** leaking out of a gas port **194**. The concentration of oxygen in the stereolithographic resin **104** is highest at the stereolithographic resin receiving surface **204**.
(26) FIGS. **5** to **9** show schematic views of one embodiment of an apparatus for making a stereolithographic object, the apparatus being generally indicated by the numeral **200**. The figures taken in sequence illustrate steps of an embodiment of a method for making a stereolithographic object utilizing vessel **100**. Coordinate axes are shown in the figures where x and y are horizontally orientated and z is vertically orientated.
(27) Apparatus **200** provides vessel **100** with photolithographic resin receiving surface **204** on which a layer of stereolithographic resin **104** can be disposed. In the context of the present document, a stereolithographic resin is a liquid that hardens when exposed to a radiation, examples of which include but are not not limited to visible and invisible light in the form of ultraviolet light, for example. A light that is capable of curing a stereolithographic resin is known as actinic light. Examples of actinic light include spectral components having a wavelength of, for example, 355 nm, 385 nm, and 405 nm. In some embodiments, radiation sources other than light may be used. For example, the radiation source may be ionizing or non-ionizing radiation.
(28) An example of a stereolithographic resin **104** that may be suitable for at least some applications may comprise a mixture of acrylate and methacrylate monomers and oligomers, photoinitiators, pigments, dyes and stabilizers such that the mixture polymerizes when exposed to suitable light. Example stereolithographic resins include but are not limited to Freeprint Ortho UV from Detax, Germany, and KZ-1860-CL from Allied PhotoPolymers, USA. Acrylate and methacrylate polymerization is inhibited by the presence of oxygen as it neutralizes free radicals which propagate the polymerization reaction. The degree of inhibition is dependent on factors including the oxygen concentration, photoinitiator concentration, monomer chemistry and molecular weight. Other photocurable materials that are inhibited by oxygen may be used as suitable, an example of which is acrylamide functionalized monomers.
(29) The vessel **100** is in the form of a shallow trough or dish for containing the stereolithographic resin **104**. The vessel **100** may have a volume sufficient to hold enough stereolithographic resin **104** to build an entire object without being replenished. Optionally, a conduit may connect the vessel **100** and a supply of the stereolithographic resin **104** to replenish the stereolithographic resin **104** as it is consumed. The oxygen permeable coating **103** forms the innermost layer of the bottom of the vessel **100**. The vessel **100** and contained stereolithographic resin **104** **104** can be removed from the apparatus and replaced with another vessel **100**, thus providing means for replacing damaged vessels or making objects from different stereolithographic resins **104**. The present vessel **100** may have significantly greater utility than a comparable apparatus having oxygen feed components such as nozzles, o-rings and seals for connecting and disconnecting pure or compressed oxygen feed lines to the vessel **100**.
(30) The embodiment of FIGS. **5** to **7** and **9** is configured such that in use the vessel **100** is horizontally orientated. The device **200** may, for example, have a chassis **130** with attached feet **132**, **133** configured to support the chassis above a surface such as a bench, and the vessel **100** is

mounted relative to the chassis so that when the chassis is so supported the base of the vessel **100** has a horizontal orientation. In other embodiments, the stereolithographic resin receiving surface **204** may be inclined at up to 45 degrees to the horizontal (that is, the surface is upwardly facing), provided that the vessel walls are sufficiently high to contain the resin **104**. Mounting brackets **152**, **154**, **156**, **158** may be used to ensure that apparatus components are maintained in their correct position and orientation relative to the chassis. Clamps **196**, **197** hold the vessel **100** against a mounting platform **510** that secures the vessel **100** against forces which may alter its position, and may serve to mount apparatus components and form a fluid-tight division between the upper and lower regions of the apparatus to prevent ingress of any spilled photohardenable fluid **104** which could damage delicate components.

(31) A radiation source in the form of a light source **116** may be activated so that it emits spatially and/or structured light **118** capable of selectively hardening areas of the stereolithographic resin **104** to form a section of the stereolithographic object. Light source **116** may, for example, incorporate a light manipulator such as an image projection system depicted in FIG. **10** and generally indicated with the numeral **159**, comprising light source **116** emitting light **162**, relay optics **163**, turning prism **164**, spatial light modulator **165** controllable by controller **168**, and projection lens **166**. Alternatively, light source **116** may be a light beam scanning apparatus depicted in FIG. **11** and generally indicated by the numeral **179**, comprising a laser source **171** emitting light **172** of wavelength of around 350 nm, for example, collimating and/or focusing optics **173**, scanning mirror **174** whose rotation is controllable in one or more axes by mirror controller **178**, optionally a second controllable mirror not shown in the figure, and optionally a projection lens **175** such as an F-Theta lens. Controller **178** can be configured to scan the mirror **174** (coordinated with a second mirror, if present) in a raster scanning mode, or alternatively in a vector scanning mode. FIG. **12** shows a second type of beam scanning apparatus generally indicated by the numeral **179** comprising a laser source **181** emitting light **182**, collimating and/or focusing optics **183**, polygon mirror **184** rotatable around an axis **185** and controllable by controller **188**, and optionally a projection lens **186** such as an F-Theta lens. As the apparatus of **179** may only scan light in the y-axis according to the coordinate system shown in FIG. **12**, the apparatus resides on a translation stage **187** which can move the apparatus in the x-direction, enabling the projected light to address locations in the x and y dimensions. The translation stage may comprise any one or more of linear motors, drive belts, stepper motors, rack and pinion arrangements, for example, or generally any suitable components arranged to provide translation. Apparatus **179** is suitable for operating in a raster scanning mode. The light source may, in some embodiments, comprise an incandescent light or light emitting diode, for example. Any suitable light source may be used.

(32) Referring again to FIGS. **5** to **9**, a positioner **120** capable of linear motion along the z-direction is coupled to and moves a platform **121** in the form of an inverted platform on which the stereolithographic object **122** being made is mounted. The positioner **120** positions the object being made **122** relative to the upwardly facing surface **103** of the vessel **100**. The positioner may comprise any one or more of linear motors, drive belts, stepper motors, rack and pinion arrangements, for example, or generally any suitable components arranged to provide linear motion.

(33) A sequence of actions can be performed with the apparatus **200** to form a new section of the stereolithographic object being made **122**. The process begins as shown in FIG. **5**, with the previous sections of the object under fabrication **122** spaced apart from the surface **204**.

(34) Next, as shown in FIG. **6**, positioner **120** lowers the object being made **122** towards the stereolithographic resin receiving surface **204**. The object **122** comes to a final position which is substantially, and generally but not necessarily a little more than one section-thickness (say, within 100% of a section thickness) above the stereolithographic resin receiving surface **204**.

(35) The thickness of one section is typically in the range of 10 micrometers to 250 micrometers,

but it may be less if particularly fine fabrication resolution is required, and greater if a relatively coarse fabrication resolution is required.

(36) Next, as shown in FIG. **7**, light **118** having spatial features in accordance with the sectional geometry of the object being made is emitted from light source **116** and is transmitted through **101** and **210** to selectively harden regions of the layer of stereolithographic resin **104** in contact with the previously formed sections **122** to form a new hardened section **124**.

(37) When the concentration of oxygen in a layer of stereolithographic resin at the stereolithographic resin receiving surface **204** is sufficiently high, the polymerization of the layer of stereolithographic resin is inhibited, resulting in a layer of unhardened stereolithographic resin **125** ("uncured stereolithographic resin") in between the hardened layer **124** and the surface **204**. FIG. **7** illustrates a highlighted region R which is shown in detail in FIG. **8** wherein the layer of uncured stereolithographic resin **125** has thickness labeled d. In practice, depending on the properties of the stereolithographic resin **104** and the vessel **100**, the thicknesses d for a range of stereolithographic resins was measured to be up to 80 μm. The presence of a layer of uncured stereolithographic resin **125** greatly facilitates the separation of the hardened layer **124** away from the surface **204** as presented in the next step.

(38) Next, as shown in FIG. **9**, mechanical actuator **120** is engaged to raise the previously formed sections **122** and newly formed section **124**. As the section **124** is pulled away from the vessel bottom **103**, the presence of a layer of uncured stereolithographic resin **125** permits ingress of further stereolithographic resin **104** into the gap to facilitate separation. This ingress of stereolithographic resin **104** prevents vacuum forces from opposing the separation of the section **124** from the vessel bottom **103** and which may generally be applied to the object. In practice a separation distance of around 200 microns may be sufficient for some embodiments. Once the final position is reached, the apparatus **200** is ready for the process to start again. Repeating this sequence of actions enables a multilaminate object to be fabricated section by section.

(39) The presence of the uncured layer of stereolithographic resin **125** significantly reduces the forces during the separation of a formed layer **124** from the vessel **100**. In configurations which have no appreciable layer of uncured stereolithographic resin **125**, the vessel **100** and newly formed layer **124** form a seal which requires greater separation force to overcome. This greater force may be destructive to delicate objects being fabricated.

(40) A drop of photohardenable resin Freeprint Ortho UV from Detax, Germany, was placed under a glass microscope slide suspended on 200 μm spacers over the experimental vessel. An irrandiance 5 mW/cm2 and wavelength 385 nm light source was used to illuminate the underside of the vessel for 4 seconds. The glass microscope slide was removed from the vessel and a spot of cured resin was attached to the glass. Vernier calipers were used to measure the thickness of the cured resin spot to be 170 microns, 30 microns less than the 200 micron space. The layer of uncured stereolithographic resin **125** in this example therefore has a thickness of 30 microns. The stereolithographic resin receiving surface **204** of the experimental vessel (and vessel **100**) has a flat configuration. A flat section of consistent thickness may subsequently be formed.

(41) In other embodiments, the stereolithographic resin receiving surface **204** may be configured such that its surface is caused to adopt a configuration other than a flat configuration (e.g. domed). This changes the distribution of the stereolithographic resin accordingly so that the spatial configuration of the stereolithographic resin between the surface **204** and the object is changed or modified to the desired shape. For example, a curved configuration may be advantageous if the object being made has a rounded shape or if the light source **116** projects light **118** to a curved focal plane. It will be understood that the configuration of the upper surface **204** may be any suitable desired geometrical arrangement.

(42) The positioner **120**, the light source **116**, and possibly other parts of the apparatus may be in communication with and may be controlled by a controller **160** to coordinate the apparatus to make the stereolithographic object **122**. These and other components may be connected by wires, cables,

wireless, or any other suitable means. In this embodiment, the controller **160** comprises a processor unit **220**, schematically illustrated in FIG. **13**. The processor unit **220** may include a suitable logic device **250** such as, or similar to, the INTEL PENTIUM or a suitably configured field programmable gate array (FPGA), connected over a bus **280** to a random access memory **240** of around 100 Mb and a non-volatile memory such as a hard disk drive **260** or solid state non-volatile memory having a capacity of around 1 Gb. The processor has input/output interfaces **270** such as a universal serial bus and a possible human machine interface **230** e.g. mouse, keyboard, display etc. Device components may be controlled using commercially available machine-to-machine interfaces such as LABVIEW software together with associated hardware recommended by the commercial interface provider installed on the processor unit **220**, over USB or RS-232 or TCP/IP links, for example. Alternatively, custom driver software may be written for improved performance together with custom printed circuit boards. Alternatively, the processor unit **220** may comprise an embedded system.

(43) In this embodiment, the controller **160** is in communication with another processor which is adapted for determining instructions and/or information for the device. In alternative embodiments, the processors are the same processor. An example of another processing unit comprises a logic device such as, or similar to, the INTEL PENTIUM or a suitably configured field programmable gate array (FPGA), connected over a bus to a random access memory of around 100 Mb and a non-volatile memory of such as a hard disk drive or solid state non-volatile memory having a capacity of around 1 Gb. Generally, the configuration may be similar or identical to that shown in FIG. **13**. The processor has a receiver such as a USB port (or Internet connection, for example) for receiving information representing a solid object, stored on a USB FLASH device, for example. The information may be encoded in a file generated by a Computer Aided Design (CAD) program, the information specifying the geometry of the object. The microprocessor runs a decomposer program implementing an algorithm that decomposes (or transforms) the information into data indicative of a plurality of sections to be formed sequentially by the device, the stereolithographic resin being used to make the solid object. The program may have been installed onto the processor from tangible media such as a DVD or USB memory stick, for example, that stored the program. In an alternative embodiment, the decomposer may be a dedicated hardware unit. A series of sections through the object are determined, each section corresponding to a solid section to be formed. The sections may then be further processed to represent the geometry of each section as a rasterised bitmap. The sections or bitmaps may then be used to control the device.

(44) It will be appreciated that embodiments may be used to make an object of generally any shape or size, including jewelry such as rings, prototype car components, micro-components for precision machines, models for investment casting, and architectural or design features for a building.

(45) Now that embodiments have been described, it will be appreciated that some embodiments may have some of the following advantages: A layer of uncured stereolithographic resin between the hardened layer and vessel bottom may be achieved without actively feeding oxygen (through concentration by pressure or purification) to the vessel. This may improve the cost and simplicity of the apparatus and may make it more reliable. The layer of uncured stereolithographic resin reduces the separation forces in which case the stereolithographic object being formed has lower risk of damage; The fabrication is faster because of improved stereolithographic resin flow and reduced separation distances being required; The vessel bottom wall ameliorates sagging of the oxygen permeable element due to the force of gravity and/or approach or separation forces. This prevents layers of inaccurate thickness or defective layers being formed which may degrade the fidelity of the fabricated stereolithographic objects, thereby improving the quality of the stereolithographic object. Some embodiments may not require an oxygen concentrator such as a pressure-swing column or external pressurized oxygen cylinder to deliver concentrated or pressurized oxygen to the vessel, nor may not require costly or inconvenient valve connections between an oxygen supply system and the vessel. The cost of the system device may be reduced.

The device may require less component connectivity making it more reliable, and the user may not need not pay attention to gas connectors when installing and removing the vessel, making it simpler and more convenient to operate.
(46) It will be appreciated that numerous variations and/or modifications may be made to the invention as shown in the specific embodiments without departing from the spirit or scope of the invention as broadly described. The present embodiments are, therefore, to be considered in all respects as illustrative and not restrictive.
(47) In the claims which follow and in the preceding description of the invention, except where the context requires otherwise due to express language or necessary implication, the word "comprise" or variations such as "comprises" or "comprising" is used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments of the invention.
(48) It is to be understood that, if any prior art publication is referred to herein, such reference does not constitute an admission that the publication forms a part of the common general knowledge in the art, in Australia or any other country.

Claims

1. A vessel for receiving a stereolithographic resin, the vessel comprising: a casting frame comprising at least one perimeter exterior wall defining a plurality of peripheral gas ports in fluid communication with an atmosphere, the plurality of gas ports extending between opposite sides of the at least one perimeter exterior wall; an oxygen permeable element comprising a cast oxygen permeable substrate cast within the casting frame and the plurality of gas ports, the oxygen permeable element comprising a stereolithographic resin receiving surface that is an interior vessel surface, the oxygen permeable substrate comprising at least one marginal surface separate from the stereolithographic resin receiving surface, wherein the at least one marginal surface is flush with an external surface of the at least one perimeter exterior wall for passive ingress of oxygen from the atmosphere and transport of the oxygen ingressed to the stereolithographic resin receiving surface to enable formation of a layer of oxygenated stereolithographic resin.
2. A vessel defined by claim 1, wherein the combined area of the at least one marginal surface is greater than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, and 32% of the area of the stereolithographic resin receiving surface.
3. A vessel defined by claim 1, wherein the combined area of the at least one marginal surface is less than at least one of 0.1%, 1%, 2%, 4%, 8%, 16%, and 32% of the area of the stereolithographic resin receiving surface.
4. A vessel defined by claim 1, wherein the oxygen permeable element has a thickness greater than at least one of 1 mm, 2 mm, 4 mm, 8 mm, 16 mm, 32 mm, 64 mm and 128 mm.
5. A vessel defined by claim 1, wherein the oxygen permeable element has a thickness less than at least one of 1 mm, 2 mm, 4 mm, 8 mm, 16 mm, 32 mm, 64 mm and 128 mm.
6. A vessel defined by claim 1, wherein the oxygen permeable element has a thickness no less than one hundredth of the square root of the area of the stereolithographic resin receiving surface.
7. A vessel defined by claim 1, wherein the oxygen permeable element has a thickness no more than one tenth of the square root of the area of the stereolithographic resin receiving surface.
8. A vessel defined by claim 1, wherein the oxygen permeable element comprises a semipermeable coating at the stereolithographic resin receiving surface.
9. A vessel defined by claim 1, wherein the oxygen permeable element comprises a cast semipermeable coating cast onto the oxygen permeable substrate.
10. A vessel defined by claim 9, wherein the cast semipermeable coating has a thickness in the range of 1 μm to 20 μm.
11. A vessel defined by claim 10, wherein the cast semipermeable coating has a thickness in the

range of 5 μm to 10 μm.
12. A vessel defined by claim 9, wherein the cast semipermeable coating has an oxygen permeability of no less than one of 10 Barrer, 30 Barrer, 1000 Barrer, 2000 Barrer and 9000 Barrer.
13. A vessel defined by claim 9, wherein the semipermeable coating has an oxygen permeability of no more than one of 20 Barrer, 50 Barrer, 2000 Barrer, 3000 Barrer and 10 000 Barrer.
14. A vessel defined by claim 1, wherein the oxygen permeable element comprises a semipermeable coating on the oxygen permeable substrate.
15. A vessel defined by claim 1 wherein an exterior vessel surface comprises the at least one marginal surface.